PAPER_845: Chandra Survey — MACS J0416 Gravitational Lensing, 3C 58, Exoplanet, SMBH, Westerlund 1

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Abstract

Five systems from the Chandra Survey Batch are analyzed: MACS J0416.1-2403 (gravitational lensing cluster), 3C 58 (young pulsar wind nebula from the 1181 AD supernova), Chandra Exoplanet Survey, SMBH Survey (12 galaxies), and Westerlund 1 (massive star cluster). Gravitational lensing amplifies observed X-ray flux but does not alter intrinsic F_U_Bi_i — the UQFF buoyancy force is a source-frame quantity.

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1. Systems and Parameters

System	M (kg)	r (m)	Description
MACS J0416.1-2403	1.989e45	1.47e23	Frontier Fields galaxy cluster
3C 58	2.8e30	3.09e19	PWN from 1181 AD SN (debated)
Exoplanet Survey	1.989e30	3.09e16	Composite stellar systems
SMBH Survey	1e40	3.09e22	12 galaxy composites
Westerlund 1	6.3e34	3.7e19	Super star cluster

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2. Gravitational Lensing and UQFF

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MACS J0416.1-2403: massive galaxy cluster (M ~ 10^15 M_sun)
Strong lensing creates multiple images of background sources.

Lensing magnification: mu = theta_E / theta_S
This amplifies OBSERVED flux but NOT intrinsic F_U_Bi_i.

F_U_Bi is computed from source-frame M, r parameters:
F_U_Bi(MACS) = -F_0 + momentum + G*M/r^2 + rho_vac + F_LENR

F_LENR = 6.17e37 N dominates at all scales.
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3. 3C 58 — Historical Supernova

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3C 58: pulsar wind nebula associated with SN 1181 (guest star)
Age attribution debated: may be older than 1181 AD
Central pulsar: PSR J0205+6449, P = 65.7 ms

M = 2.8e30 kg (neutron star mass)
r = 3.09e19 m (PWN extent)
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F_U_Bi ~ 2.65e208 N (positive buoyancy)
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4. Westerlund 1

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Most massive young star cluster in the Milky Way
M_cluster ~ 6.3e4 M_sun, r ~ 1.2 pc
Contains: magnetars, Wolf-Rayet stars, OB supergiants

F_U_Bi ~ 2.65e208 N (positive buoyancy)
F_LENR dominates cluster-scale physics
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Conclusion

The Chandra Survey batch establishes that gravitational lensing (MACS J0416) amplifies observed signal without modifying UQFF intrinsic forces. All five systems show positive buoyancy with F_LENR dominance. Westerlund 1 as the Milky Way's most massive young cluster provides a rich testbed for UQFF stellar-scale predictions.
